

Imaging Information for Uterine Leiomyosarcoma

Quick Tips – Care Guide



Uterine sarcomas are rare and heterogeneous malignant mesenchymal tumors, representing only 1% to 3% of gynecological cancers and 3% to 7% of uterine malignancies, with an annual global incidence of 0.5 to 3.3 cases per 100,000 women. These tumors are the most lethal among uterine malignancies, exhibiting a 5-year overall survival rate of 25% to 70%, depending on histologic subtype and stage.^{1, 2, 3, 4, 5} Uterine leiomyosarcoma is the predominant subtype (60%-70%). [Learn more.](#)

Overview

PRIMARY TREATMENT

Management of uterine leiomyosarcoma is challenging because evidence is limited and sometimes conflicting. For resectable disease, prompt and complete surgical resection is the mainstay of treatment and remains the only curative option. For unresectable, metastatic, or recurrent disease, treatment is typically multimodal and may include surgery, chemotherapy, and radiotherapy. Care is best delivered in specialized referral centers.

FOLLOW-UP

Because uterine leiomyosarcoma is aggressive and carries a high risk of recurrence, close clinical and radiologic follow-up is essential. Patients should be informed about symptoms of recurrence and the importance of seeking prompt evaluation. Follow-up with history and physical examination is recommended every 3–4 months for the first 2–3 years, then every 6–12 months, with contrast-enhanced CT of the chest, abdomen, and pelvis as indicated.

TREATMENT OF RECURRENT DISEASE

Recurrent disease commonly involves the abdomen, pelvis, or lungs. Management should be individualized using a multidisciplinary approach similar to that for advanced or metastatic disease. Treatment may include systemic chemotherapy and/or locoregional therapies, depending on the site of recurrence, resectability, prior treatment, and disease-free interval.

NOVEL THERAPEUTIC AGENTS AND FUTURE PERSPECTIVES

Advances in molecular profiling and multi-omics may support more personalized treatment strategies in uterine leiomyosarcoma. Investigational approaches include anti-angiogenic agents, immune checkpoint inhibitors, HER2-targeted therapies, PARP inhibitors, and combination regimens. Progress in clinical research remains limited by the rarity of the disease.

Evaluation

Leiomyomas typically appear on ultrasound as well-circumscribed, homogeneous, hypoechoic masses with variable vascularity. However, their sonographic appearance can overlap with that of leiomyosarcoma (LMS).

In earlier studies, LMS was associated with several ultrasound features, including large size (greater than 8 cm), irregular borders, cystic or necrotic areas, increased central and peripheral vascularity, higher peak systolic velocity, lower resistive index, and rapid growth. In one small study involving eight LMS cases, 21 cellular leiomyomas, three smooth muscle tumors of uncertain malignant potential (STUMPs), and 225 benign leiomyomas, increased central and peripheral vascularity showed a sensitivity of 100% and a specificity of 86% for diagnosing LMS. However, the positive predictive value was only 19%.

Overall, the diagnostic performance of ultrasound remains uncertain. Another small study of 111 patients—including six with LMS, seven with carcinosarcoma, and 98 with leiomyoma—found no meaningful difference in ultrasound appearance, including mean resistive index, between LMS and leiomyomas.

MRI Evaluation

Among imaging modalities, MRI provides the highest accuracy for characterizing uterine masses before intervention because of its superior soft-tissue contrast, larger field of view, diffusion-weighted imaging, and multiplanar capabilities. For procedural planning, MRI also allows more precise localization of a lesion within the uterus.

Using MRI to map and characterize symptomatic leiomyomas before treatment has been shown to change the management approach in up to 20% of patients. In the preprocedural setting, MRI features have also been studied for their ability to distinguish leiomyosarcoma (LMS) from leiomyomas and atypical leiomyomas.

CT and PET/CT Evaluation

CT is not typically used for the diagnosis or local staging of uterine leiomyosarcoma (LMS). Fluorine-18 fluorodeoxyglucose (18F-FDG) PET/CT has only a limited role in detecting LMS. Although uterine leiomyomas usually demonstrate mild 18F-FDG uptake, they can occasionally show intense uptake, especially in premenopausal women. This increased uptake may be related to factors such as hormonal influences, cellularity, vascularity (including microvessel density), tumor cell proliferation, inflammatory cells, and the presence of endometrial tissue.

LMS is the **most common subtype of uterine sarcoma**, followed by **endometrial stromal sarcoma (ESS)**, **undifferentiated endometrial sarcoma**, and **adenosarcoma**. **Carcinosarcoma** (previously called mixed malignant müllerian tumor) is no longer classified as a sarcoma subtype. In 2009, the International Federation of Gynecology and Obstetrics reclassified uterine carcinosarcoma as a variant of endometrial carcinoma. As a result, it should not be included as a sarcoma subtype in current sarcoma studies.

Among uterine sarcomas, LMS is the subtype that most often presents as a **myometrial-based mass** and is therefore the one most likely to mimic a benign leiomyoma. In contrast, ESS arises from the **endometrial stroma**. About 90% of ESS cases present with **abnormal uterine bleeding**, and nearly all involve the endometrium. Only rarely does ESS appear as an isolated myometrial mass without associated endometrial thickening.

For this reason, **deep endometrial curettage** can be helpful in the preoperative diagnosis of ESS. By contrast, LMS does not usually involve the endometrium and therefore is not typically detected through endometrial sampling.

Resources

[National LMS Foundation: Uterine LMS](#)

[Memorial Sloan Kettering: Nomograms for Uterine Leiomyosarcoma](#)

[Predict Postresection 5-year Overall Survival for Patients with Uterine LMS](#)

[Uterine Fibroids – What You Should Know Before Having Fibroid Surgery](#)

[Endometrial Sampling for Preoperative Diagnosis of Uterine LMS](#)

[Uterine Sarcoma Treatment \(PDQ®\)](#)

[Surgical Staging of Uterine Sarcoma](#)

[Laparoscopic Power Morcellators](#)

[MRI Screening of Uterine LMS](#)

[Preoperative Blood Inflammatory Markers](#)



Disclaimer: The content is provided for general informational purposes only and is not intended as, nor should it be considered a substitute for, professional medical advice. Do not use the information on this website for diagnosing or treating any medical or health condition. If you have or suspect you have a medical problem, promptly contact your professional healthcare provider.



Sarcoma & LMS Pathways

Care Guide - Quick Tips for Patients & Families

Trusted guidance for navigating diagnosis, treatment, and leiomyosarcoma

www.NLMSF.org